# Chapter 5: Electrostatics Fields in Material

The following subjects will be discussed

- Polarization in Dielectrics.
- Current in Conductors.
- Conservation of Charge Equation.
- Electrostatic Boundary Conditions.

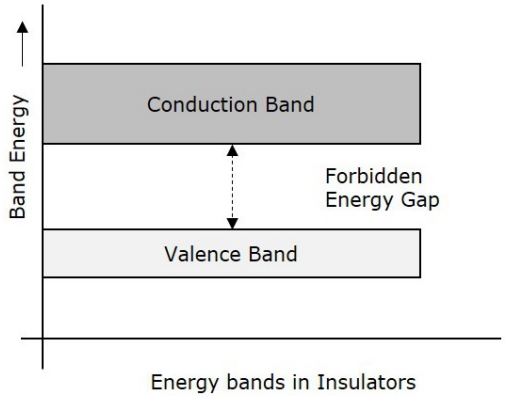
# Electrostatic Fields in Material

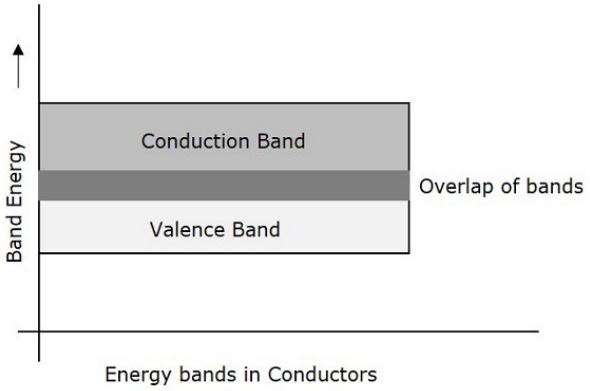
Materials can be classified based on the effect of external electrostatic fields as dielectrics (insulators), semi-conductors and conductors.

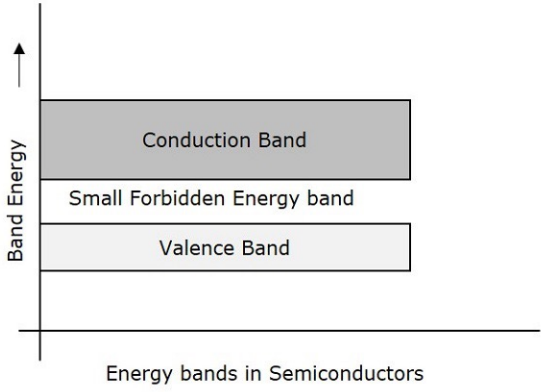
Dielectrics are characterized by the lack of free charges; instead, the charges are bound to their atoms. Their energy band structure does not contain low-energy states available for conduction.

Conductors are those materials that naturally possess abundant vacant energy states in their energy band structure at very low energy values, so that free-charges exist in big numbers ready for conduction.

Semi-conductors fall between the two groups of material.

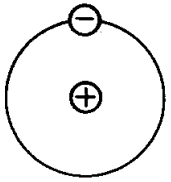


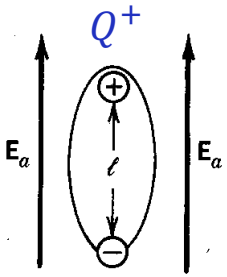



# The Electric Polarization

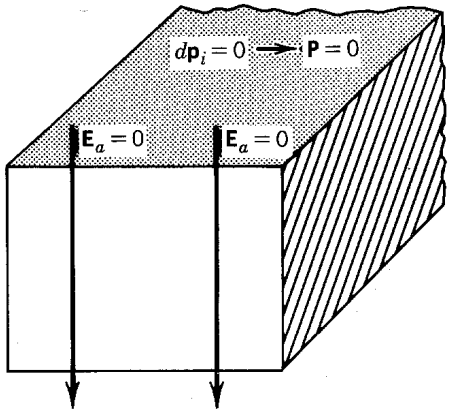
When dielectrics are subjected to external electrostatic fields, they become polarized.



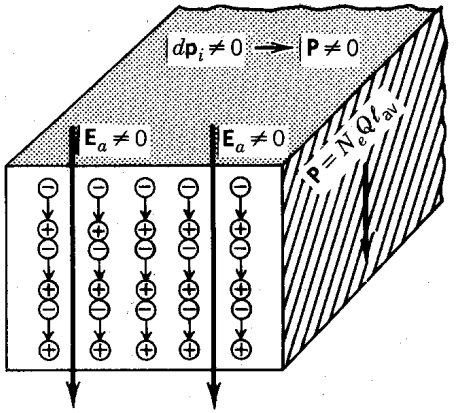
Atom with no external field




Atom with external field




Dielectric with no external field




Dielectric with external field

With external electrostatic field applied, dipoles are created all over the dielectric material, each with dipole moment of $\vec{p} = Q\vec{d}$ (see next slide).

Over some incremental volume $\Delta v$, we define the electric Polarization vector as,

$$\vec{P} = \lim_{\Delta v \to 0} \frac{\sum Q_i \vec{d}_i}{\Delta v} \qquad (C/m^2)$$

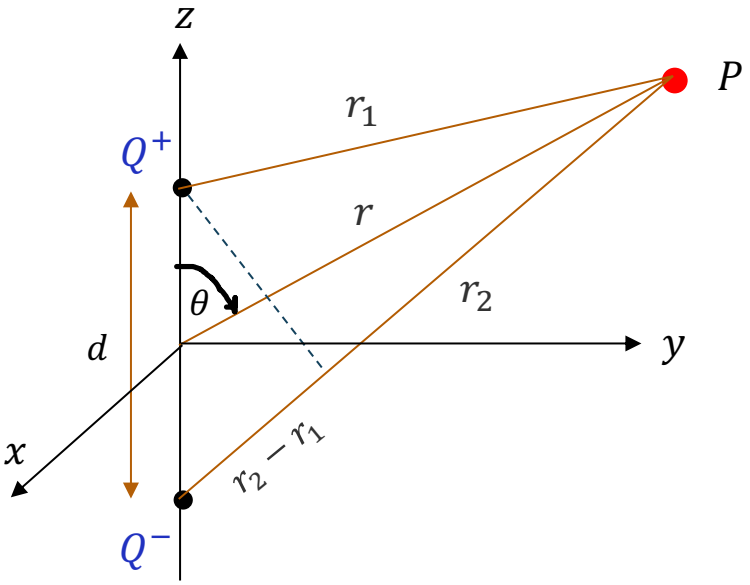
This polarization vector is proportional to the applied external field. $\vec{P} \propto \vec{E}$

# Electric Field Due to a Dipole

The diploe problem is an interesting one because it is encountered in many subjects in science and engineering (e.g., polarized atom).

The electric potential at any point $P$ is written as,

$$V = \frac{Q}{4\pi\varepsilon}\frac{1}{r_1} - \frac{Q}{4\pi\varepsilon}\frac{1}{r_2} = \frac{Q}{4\pi\varepsilon}\left[\frac{r_2 - r_1}{r_1\, r_2}\right]$$




Far away from the dipole, $r \gg d$,

$$r_1\, r_2 = r^2\,, \qquad r_2 - r_1 \approx d\cos\theta$$

$$V = \frac{Q}{4\pi\varepsilon}\left[\frac{d\cos\theta}{r^2}\right]$$

We define the dipole "moment" as, $\vec{p} = Q\vec{d}$ so that $d\cos\theta = \vec{d}\cdot\vec{a}_r$

$$V = \frac{\vec{p}\cdot\vec{a}_r}{4\pi\varepsilon\, r^2}$$

$$\vec{E} = -\vec{\nabla}V = \frac{|\vec{p}|}{4\pi\varepsilon r^3}\left(2\cos\theta\,\vec{a}_r + \sin\theta\,\vec{a}_\theta\right)$$

# The Electric Polarization

To find the potential at a point $P$ due to an incremental volume $\Delta v$ containing many individual dipoles, we integrate the previous result.

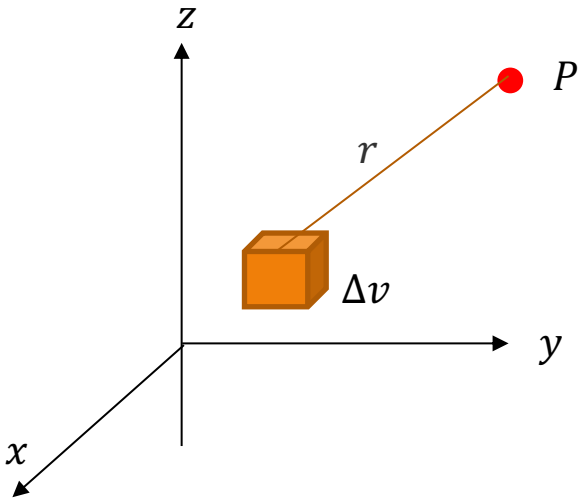
$$dV = \frac{\vec{P} \cdot \vec{a}_r}{4\pi\varepsilon\, r^2}\Delta v$$

The result is of the form,

$$V = \int_S \frac{\vec{P} \cdot \overrightarrow{dS}}{4\pi\varepsilon\, r} + \int_v \frac{-\vec{\nabla} \cdot \vec{P}}{4\pi\varepsilon r}\, dv$$

Potential due to a surface charge distribution

Potential due to a volume charge distribution




So, we can represent the effect of polarization as,

a polarization surface charge distribution $\rho_{sp} = \vec{P} \cdot \vec{a}_n$

a polarization volume charge distribution $\rho_{vp} = -\vec{\nabla} \cdot \vec{P}$

These charges are bound charges (not free charges) exist only because of the external field. Also,

$$\int_S \rho_{sp}\, dS + \int_v \rho_{vp}\, dv = 0$$

# The Electric Flux Density

Previously, we had the following equations for the electric flux density.

$$\vec{D} = \varepsilon_o \vec{E} \qquad \text{and} \qquad \vec{\nabla} \cdot \vec{D} = \rho_v$$

where $\rho_v$ is the free charge density. Inside a dielectric material, the electric polarization needs to be accounted for by adding the polarization bound charge density. The equation above is changed to,

$$\rho_{total} = \rho_v + \rho_{vp} = \vec{\nabla} \cdot \varepsilon_o \vec{E}$$

Using the relation $\rho_{vp} = -\vec{\nabla} \cdot \vec{P} \quad \rightarrow \quad \vec{\nabla} \cdot \vec{D} = \vec{\nabla} \cdot \varepsilon_o \vec{E} + \vec{\nabla} \cdot \vec{P} = \vec{\nabla} \cdot \left(\varepsilon_o \vec{E} + \vec{P}\right)$

Electric Flux Density inside a dielectric is given by, $\vec{D} = \varepsilon_o \vec{E} + \vec{P}$ $(C/m^2)$

The electric flux inside the dielectric is increased due to polarization.

As the polarization is proportional to the applied external field $(\vec{P} \propto \vec{E})$, we can write

$$\vec{P} = \varepsilon_o \gamma_e \vec{E}$$

$\gamma_e$ : Electric susceptibility

Electric Flux Density becomes, $\vec{D} = \varepsilon_o (1 + \gamma_e) \vec{E} = \varepsilon_o \varepsilon_r \vec{E} = \varepsilon \vec{E}$

$\varepsilon_r = 1 + \gamma_e$ is the relative permittivity, or dielectric constant (unitless)

The dielectric constant is a property of the dielectric material.

# The Static Dielectric Constant

The static dielectric constants of some common materials are given the table below.

| Material | Dielectric Constant $\kappa$ | Dielectric Strength[a] (V/m) |
|---|---|---|
| Air (dry) | 1.000 59 | $3 \times 10^6$ |
| Bakelite | 4.9 | $24 \times 10^6$ |
| Fused quartz | 3.78 | $8 \times 10^6$ |
| Neoprene rubber | 6.7 | $12 \times 10^6$ |
| Nylon | 3.4 | $14 \times 10^6$ |
| Paper | 3.7 | $16 \times 10^6$ |
| Polystyrene | 2.56 | $24 \times 10^6$ |
| Polyvinyl chloride | 3.4 | $40 \times 10^6$ |
| Porcelain | 6 | $12 \times 10^6$ |
| Pyrex glass | 5.6 | $14 \times 10^6$ |
| Silicone oil | 2.5 | $15 \times 10^6$ |
| Strontium titanate | 233 | $8 \times 10^6$ |
| Teflon | 2.1 | $60 \times 10^6$ |
| Vacuum | 1.000 00 | — |
| Water | 80 | — |

# Examples

(1) The static dielectric constant of water is 81. Find the flux density and polarization inside water if an electrostatic field of magnitude 1 V/m is applied.

$\varepsilon_{r_water} = 81$

$$\vec{D} = \varepsilon_o \varepsilon_r \vec{E} \rightarrow |D| = 81\varepsilon_o|E| = 0.717 \quad (\text{nC/m}^2)$$

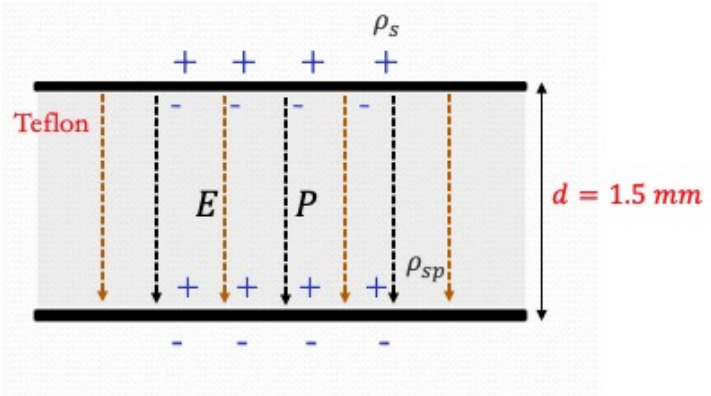
$$\vec{P} = \varepsilon_o \gamma_e \vec{E} \rightarrow |P| = (81-1)\varepsilon_o|E| = 0.707 \quad (\text{nC/m}^2)$$

(2) A slab of polystyrene of thickness 1.5 mm is subjected to an electric field of amplitude 10 kV/m across its thickness. Find the free and bound surface charge densities.

$\varepsilon_{r_polystyrene} = 2.55$

$$\rho_s = \vec{D} \cdot \vec{a}_n = |D| = 2.55\ \varepsilon_o|E| = 225.4 \quad (\text{nC/m}^2)$$

$$\rho_{sp} = \vec{P} \cdot \vec{a}_n = |P| = (2.55-1)\varepsilon_o|E| = 137 \quad (\text{nC/m}^2)$$



# Classification of Dielectric Materials

Dielectrics can be classified based on the nature of the dielectric constant, as follows.

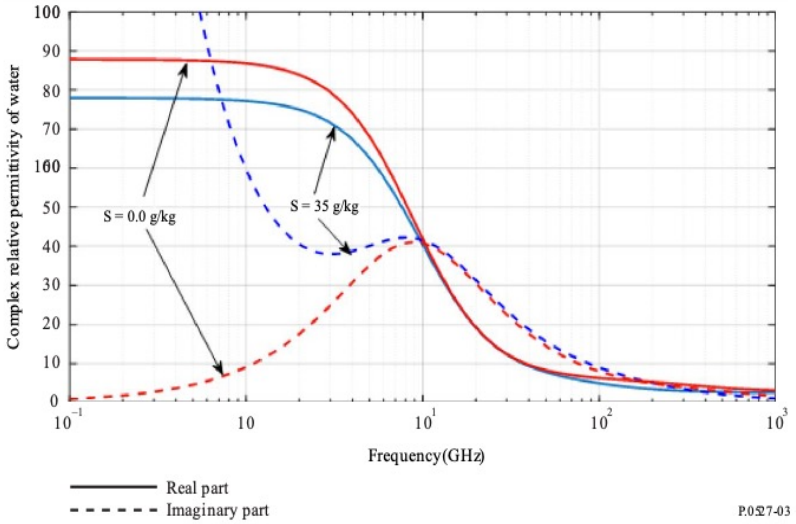
(1) Homogeneous: The value of dielectric constant is the same all over the material; it is independent of position within the material. (otherwise, it is non-homogeneous)

(2) Isotropic: The value of dielectric constant is same along all directions inside the material; it is independent of direction. (otherwise, it is non-isotropic)

(3) lossless: The value of dielectric constant is a real number. (otherwise, it is lossy, and the dielectric constant is a complex number)

(4) Non-dispersive: The value of dielectric constant is independent of frequency. (otherwise, it is dispersive. Example, sea water)

(5) Linear: The value of dielectric constant is independent of magnitude of applied electric field. (otherwise, it is non-linear)



# Field Equations in Material

All the equations studied so far are changed to include the dielectric constant of the material.

$$\vec{F} = \frac{1}{4\pi\varepsilon_o\varepsilon_r}\frac{Q\,q}{r_{12}^2}\,\vec{a}_{r_{12}} \qquad \vec{E}_p = \frac{1}{4\pi\varepsilon_o\varepsilon_r}\frac{Q}{r_p^2}\,\vec{a}_r$$

$$Q_{enclosed} = \oint \vec{D}\cdot\overrightarrow{dS} = \varepsilon_o\varepsilon_r \oint \vec{E}\cdot\overrightarrow{dS}$$

$$V = \frac{q}{4\pi\varepsilon_o\varepsilon_r r}$$

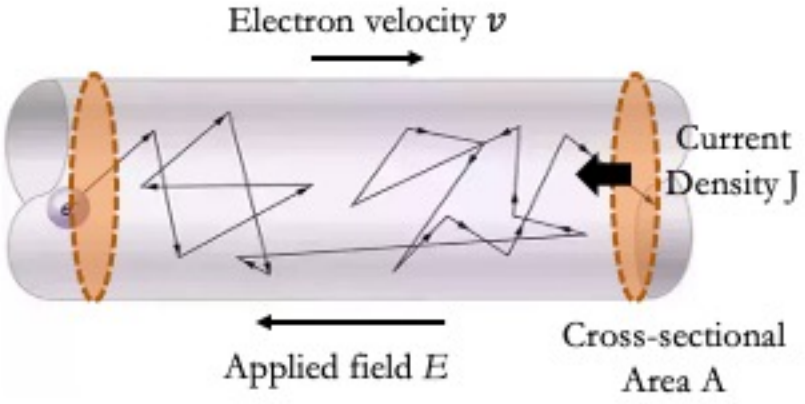
$$W_E = \frac{\varepsilon_o\varepsilon_r}{2}\int |\vec{E}|^2 dv \qquad w_E = \frac{\varepsilon_o\varepsilon_r}{2}|\vec{E}|^2$$

# Current in Conductors

Conducting materials have free electrons (loosely attached to atoms).

Conducting materials have available energy levels for conduction (paths for electrons).

When conductors are subjected to external electrostatic fields, current is generated.




With external electrostatic field applied, free charges are transported across the material, with some average velocity $\vec{v}$.

Over some incremental surface $\Delta S$, we define the electric Current Density vector as,

$$\vec{J} = \rho_v \vec{v} \qquad (\mathrm{A/m^2})$$

This current density vector is proportional to the applied external field. $\vec{J} \propto \vec{E}$

# Current in Conductors

This current density vector is proportional to the applied external field. $\vec{J} \propto \vec{E}$

Using the relation $\vec{v} = \mu\vec{E}$ , we write

$\mu$: Charge mobility ($m^2/V.s$)

$$\vec{J} = \rho_v \mu \vec{E} = \sigma\vec{E} \quad (A/m^2)$$

$\sigma$: Material conductivity ($S/m$)

The total current through the material is found by integrating the current density over the cross-sectional area.

$$I = \int_S \vec{J} \cdot \overrightarrow{dS} \quad (A)$$

We can define the resistance (using Ohm's law) as,

$$R = \frac{V}{I} = \frac{\int_L \vec{E} \cdot \overrightarrow{dl}}{\sigma \int_S \vec{E} \cdot \overrightarrow{dS}}$$

The associated power can be written as,

$$P = VI = \int_L \vec{E} \cdot \overrightarrow{dl} \int_S \vec{J} \cdot \overrightarrow{dS} = \int_v (\vec{E} \cdot \vec{J})\, dv \quad (W)$$

Power density is defined as

$$w_P = (\vec{E} \cdot \vec{J})$$

($W/m^3$)

# Examples

(1) In response to some applied electrostatic field, the current density vector inside a conducting material is given by,

$$\vec{J} = \frac{1}{r^3}\left(2\cos\theta\,\vec{a}_r + \sin\theta\,\vec{a}_\theta\right) \quad (\mathrm{A/m^2})$$

Find the total current through a spherical shell of radius 10 cm. $\overrightarrow{dS} = r^2 \sin\theta\, d\theta\, d\varphi\, \vec{a}_r$

$$I = \int_S \vec{J}\cdot\overrightarrow{dS} = \frac{2}{r}\int_0^{2\pi} d\varphi \int_0^{\pi} \cos\theta \sin\theta\, d\theta = \frac{2}{0.1}(2\pi)(0) = 0$$

(2) The current density vector inside a conducting material is given by,

$$\vec{J} = \frac{e^{-10^3 t}}{\rho^2}\vec{a}_\rho \quad (\mathrm{A/m^2})$$

Find the total current through the surface $\rho$=2 and $0 \le z \le 3$, at $t$=0. $\overrightarrow{dS} = \rho\, d\varphi\, dz\, \vec{a}_\rho$

$$I = \int_S \vec{J}\cdot\overrightarrow{dS} = \frac{e^{-10^3 t}}{\rho}\int_0^{2\pi} d\varphi \int_0^{3} dz = \frac{1}{2}(2\pi)(3) = 3\pi \quad (\mathrm{A})$$

# Conservation of Charge
# The Continuity Equation

The rate of change (positive/negative) of free charges within a closed surface S is equal to the amount of current entering/leaving that surface.

$$\int_v \frac{\partial}{\partial t}\rho_v \, dv = -\oint \vec{J}\cdot\overrightarrow{dS} = \int_v \vec{\nabla}\cdot\vec{J} \, dv \quad \rightarrow \quad \vec{\nabla}\cdot\vec{J} + \frac{\partial}{\partial t}\rho_v = 0$$

Conservation of charge

For DC currents, $\frac{\partial}{\partial t}\rho_v = 0 \quad \rightarrow \quad \vec{\nabla}\cdot\vec{J} = 0$

The continuity equation allows as to study the relaxation of charges. Consider an initial charge density ($\rho_{vo}$) situated at a certain location inside a material. How fast will these charges disperse/relax (because of Coulombic forces)?

Starting with the continuity equation, substitute $\vec{J} = \sigma\vec{E}$ $\vec{D} = \varepsilon\,\vec{E}$ $\vec{\nabla}\cdot\vec{D} = \rho_v$

to obtain the following equation to obtain a differentia equation $\frac{\partial\rho_v}{\partial t} + \frac{\sigma}{\varepsilon}\rho_v = 0$

which has the following solution $\rho_v(t) = \rho_{vo}e^{-(\sigma/\varepsilon)t}$

$\frac{\varepsilon}{\sigma}$ : Relaxation time

Copper: $10^{-19}$ s

Quartz: 50 Days

# Example

In response to some applied electrostatic field, the current density vector inside a conducting material is given by,

$$\vec{J} = \frac{e^{-10^3 t}}{\rho^2}\vec{a}_\rho \quad (\mathrm{A/m^2})$$

Find the free charge volume density within the surface $\rho$=2 and $0 \le z \le 3$, at $t$=10 ms. Assume $\rho_v(\infty)$ =0.

$$\frac{\partial}{\partial t}\rho_v = -\vec{\nabla}\cdot\vec{J} = -\frac{1}{\rho}\frac{\partial}{\partial \rho}\left(\rho\frac{e^{-10^3 t}}{\rho^2}\right) = \frac{e^{-10^3 t}}{\rho^3}$$

$$\rho_v(t) = \int \frac{e^{-10^3 t}}{\rho^3}\,dt = (-10^3)\,\frac{e^{-10^3 t}}{\rho^3} + C$$

$$\rho_v(\infty) = 0 \;\rightarrow C = 0$$

$$\rho_v(10\,ms) = (-10^3)\,\frac{e^{-10^3(10^{-2})}}{2^3} = -5.67 \quad (\mathrm{nC/m^3})$$

# Convection Current

Dielectric materials can experience material breakdown when subjected to strong electric fields. The atoms become energetic to the extent of disintegration, giving rise to convection current. Example, lightning: the breakdown of the atmosphere.

Dielectric strength: the maximum field a material can stand before breakdown occurs.

| Material | $E_{cr}$ (MV/m) | Material | $E_{cr}$ (MV/m) |
|---|---|---|---|
| Air (atmospheric pressure) | 3 | Bakelite | 25 |
| Barium titanate ($BaTiO_3$) | 7.5 | Glass (plate) | 30 |
| Freon | $\sim 8$ | Paraffin | $\sim 30$ |
| Germanium | $\sim 10$ | Silicon (Si) | $\sim 30$ |
| Wood (douglas fir) | $\sim 10$ | Alumina | $\sim 35$ |
| Porcelain | 11 | Gallium arsenide | $\sim 40$ |
| Oil (mineral) | 15 | Polyethylene | 47 |
| Paper (impregnated) | 15 | Mica | 200 |
| Polystyrene | 20 | Fused quartz ($SiO_2$) | $\sim 1000$ |
| Teflon | 20 | Silicon nitride ($Si_3N_4$) | $\sim 1000$ |
| Rubber (hard) | 25 | Vacuum | $\infty$ |

# Q & A

Send me an email if you have any questions.